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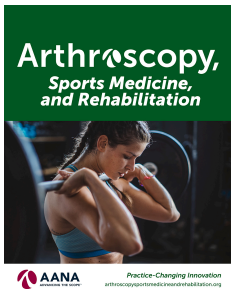

Full Return to Sport After Suture-Button Fixation for Syndesmotic Injuries in All Athletes; However, Faster Recovery Was Seen in National Collegiate Athletic Association Football Players Compared with National Football League Players

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Full Return to Sport After Suture-Button Fixation for Syndesmotic Injuries in All Athletes; However, Faster Recovery Was Seen in National Collegiate Athletic Association Football Players Compared with National Football League Players

Running Title: RTS Following Suture-Button Fixation

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Full Return to Sport After Suture-Button Fixation for Syndesmotic Injuries in All Athletes; However, Faster Recovery Was Seen in National Collegiate Athletic Association Football Players Compared with National Football League Players

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Abstract

Purpose: The purpose of this study was to determine the rate and time to return to sport (RTS) for NCAA Football and NFL players following suture-button fixation for the treatment of unstable tibiofibular syndesmotic injuries and to identify possible risk factors associated with a delay in RTS.

Methods: NCAA and NFL players who underwent suture-button fixation for an isolated syndesmotic ankle injury between 2017 and 2023 were evaluated. All player information and data were retrieved from publicly available internet sources. Information was gathered on each player's career, NFL participation, number of seasons played before and after surgery, number of games played before and after surgery, and time between surgery and RTS. RTS is defined as time until first game after syndesmotic suture-button fixation.

Results: Seven NCAA and fourteen NFL players (21 ankles) were included. The mean age at surgery was 24.90 ± 4.66 years (range, 18-34 years). A total of 100% of patients were able to RTS at their respective level of sport. The mean time between surgery and the first game played was 139.4 ± 123.6 days (range, 12 - 427 days). Players played in 23.48 ± 21.9 games after surgery (range, 3-72). NCAA players were able to RTS significantly earlier than NFL players (39.0 ± 20.4 vs 189.6 ± 123.2 days, respectively, P = 0.005, with a p-value of 0.05 denoting significance). There was no significant difference in workload following surgery for both NCAA and NFL players.

Conclusion: In this study, we found that athletes, regardless of whether they were NCAA or NFL players, were able to successfully return to their respective sports after ankle surgery. While

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NCAA players returned to play significantly faster, both groups were able to achieve similar performance levels postoperatively in terms of the number of games played.

Level of evidence: Level III, retrospective cohort study

Introduction

Unstable tibiofibular syndesmotic injuries, commonly referred to as high ankle sprains, present a significant challenge in both diagnosis and treatment, particularly among athletes[1]. These injuries involve the disruption of the ligamentous structures between the tibia and fibula, which are critical for maintaining ankle stability[2,3]. The distal tibiofibular syndesmosis is stabilized by four key ligamentous structures: the anterior-inferior tibiofibular ligament (AITFL), which resists external rotation and anterior displacement of the fibula; the posterior-inferior tibiofibular ligament (PITFL), which resists posterior displacement; the interosseous ligament (IOL), a thickened portion of the interosseous membrane providing central stability; and the inferior transverse ligament, which reinforces posterior ankle stability. Collectively, these ligaments maintain syndesmotic integrity and ankle mortise stability during weight-bearing and rotational stress. Syndesmotic stability is conferred by both osseous and ligamentous relationships between the distal tibia and fibula[2,3]. The fibula rests within a concave triangular groove (i.e., incisura) in the tibia, with its apex 6 to 8 cm proximal to the talocrural joint[2,3]. The anterior-inferior tibiofibular ligament (AITFL), interosseous ligament (IOL), interosseous membrane, posterior-inferior tibiofibular ligament (PITFL), and inferior transverse ligament compose the syndesmosis[2]. These injuries occur most commonly in sports that involve sudden twisting, cutting, or pivoting movements, especially those that put excessive rotational stress on the ankle. The most common sports include Men's football (2.42/10,000 AEs), wrestling

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(2.11/10,000 AEs), and ice hockey (1.19/10,000 AEs) had the highest high ankle sprain injury rates [4]. Left untreated or inadequately addressed, such injuries can lead to chronic pain, instability, and a prolonged absence from athletic activities.

Historically, the management of syndesmotic injuries has involved various surgical techniques, including screw fixation[5]. However, screw fixation often necessitates a secondary procedure for removal and may limit physiological movement between the tibia and fibula, potentially impacting functional recovery[5–7]. In recent years, suture-button fixation has gained traction as an alternative to traditional screw fixation, with some studies showing improved outcomes compared to syndesmotic screws such as grater fibular freedom and significantly lower hardware failure rates that could result in a secondary procedure to remove failed hardware[7–9]. This technique employs flexible sutures anchored by small buttons, allowing for dynamic stabilization of the syndesmosis while permitting natural motion between the tibia and fibula[1,8,10]. Anatomic fixation directed from the posterior cortex of the fibula to the anterolateral edge of the tibia allows for dynamic stabilization[11].

The advantages of suture-button fixation include the elimination of the need for hardware removal, reduced risk of overtightening, and potentially faster rehabilitation timelines[12,13]. As a result, it has become increasingly popular, particularly among high-performance athletes who require a reliable and efficient return to sport (RTS)[14,15]. Despite its growing acceptance, the body of literature evaluating the outcomes of suture-button fixation remains limited regarding RTS among National Collegiate Athletics Association (NCAA) football and National Football League (NFL) players. Understanding how this method performs under the extreme physical demands and performance expectations of NCAA and NFL football is both clinically and practically important.

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Therefore, the purpose of this study was to determine the rate and time to return to sport for NCAA and NFL players following suture-button fixation for the treatment of unstable tibiofibular syndesmotic injuries and to identify possible risk factors associated with a delay in RTS. return to sport We hypothesize that all NFL and NCAA players treated with suture-button fixation for unstable tibiofibular syndesmotic injuries will return to sport at their respective level of competition and return to sport will not be significantly different NFL and NCAA football players following surgery.

Methods

This retrospective cohort study included NCAA football and NFL players who sustained in-season syndesmotic ankle injuries between 2017-2023. Players were included in the study if they were Players who sustained a syndesmotic ankle injury were identified by means of publicly available sources such as NFL.com, player profiles and team news reports, and injury reports using methods like those used in previously published investigations[16–18]. Each reported injury was cross-referenced with at least 2 of the above sources to confirm diagnosis of syndesmotic ankle injury and subsequent suture-button fixation. The above sources were used to further confirm the dates of injury by cross-referencing with statistical pages to verify a gap in playing time during the time away from sport and to verify the documented date of injury and date of RTS for each player. Players were included only if they had participated in an NCAA football or NFL game before their injury. Players were excluded if their injury or surgical treatment could not be verified, if they had not participated in a game prior to the injury, did not return to NCAA or NFL after treatment, or if there was insufficient publicly available information to confirm their inclusion.

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Descriptive data for each player, including body mass index (BMI), age, and position at the time of injury, were collected from team statistical pages. Successful RTS was defined as any documented NCAA football or NFL game play after syndesmotic suture-button fixation. The total number of seasons and games played after the injury was documented. Overall career data before and after the date of injury were collected to better assess longitudinal changes in level of function.

Statistical Analysis

Descriptive statistics, including means and standard deviations (SDs), were used to summarize continuous variables such as age, BMI, time to RTS, pre-injury games played, and post-injury games played. Categorical variables, such as playing position, were summarized using frequencies and percentages. Differences between groups (e.g., by RTS) were analyzed using appropriate statistical tests such as Chi-square and Fisher's exact tests for categorical data (e.g., comparison of player position) and t-tests for continuous data (e.g., Age at time of injury or BMI), respectively, with significance set at $p < 0.05$.

Figure 1. Patient Selection Flowchart for Inclusion of NCAA and NFL Football Players Undergoing Suture-Button Fixation for Syndesmotic Ankle Injuries.

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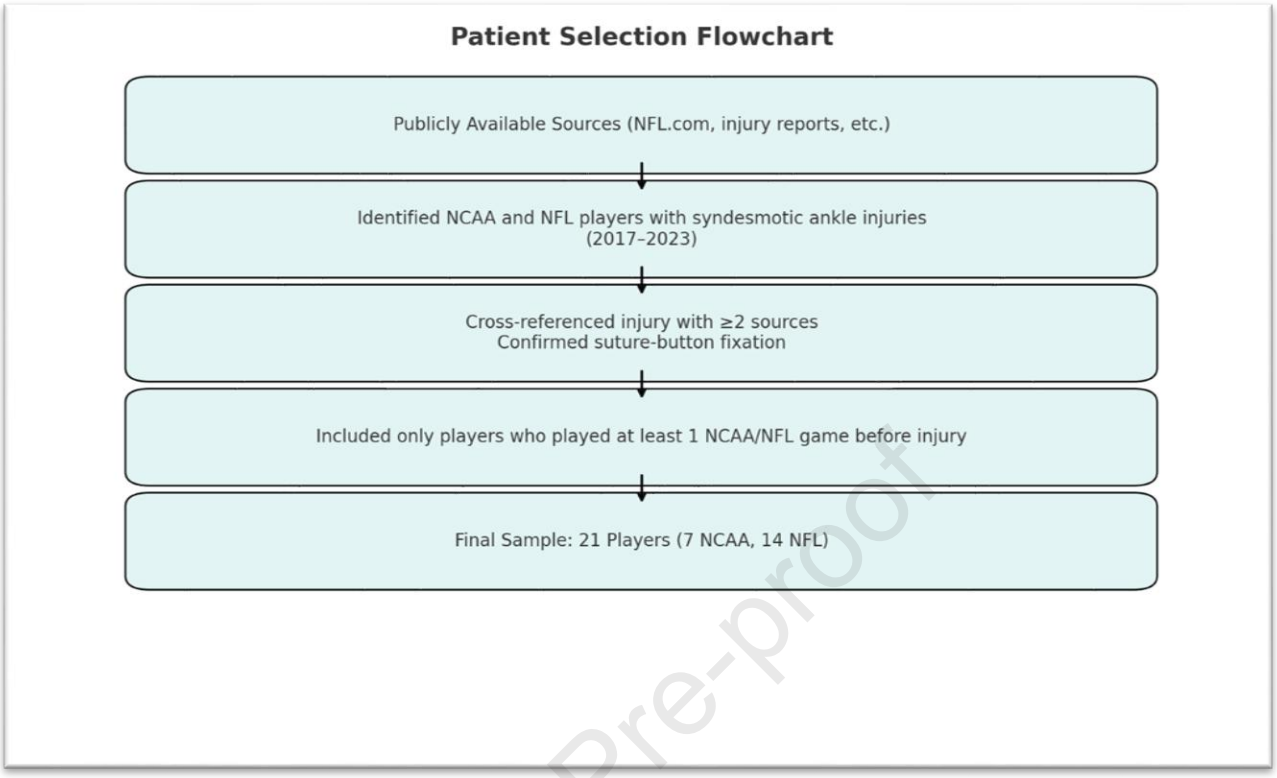

Results

Seven NCAA football and fourteen NFL players (21 ankles) were included (Figure 1). The mean age at the time of surgery was 24.90 ± 4.66 years, with arange of 18 to 34 years. There was a significant difference (p<0.001) between mean age of NCAA football players (20.0 ± 1.2) and NFL players (27.3 ± 3.6) (Table 1). There was no significant differences between the two cohorts when comparing BMI and position played. There was a significant difference (p<0.001) between pre-injury games played in the NCAA cohort (8.6 ± 1.7) and NFL cohort (13.5 ± 2.0).

Table 1. Descriptive Data for National Collegiate Athletics Association (NCAA) Football and National Football League (NFL) Players.

Variable	NCAA (n = 7)	NFL (n=14)	P Value
Age at time of injury, y, mean ± SD	20.0 ± 1.2	27.3 ± 3.6	<0.001
Body mass index at time of injury, mean ± SD	29.0 ± 4.3	29.0 ± 4.0	0.99
Position, n (%)			
Quarterback	2 (28)	2 (16)	0.57

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Running Back	0 (0)	3 (23)	0.52
Wide Receiver	2 (28)	5 (38)	0.74
Tight End	1 (14)	0 (0)	0.14
Offensive Line	1 (14)	1 (7)	0.59
Defensive Line	0 (0)	2 (16)	0.29
Cornerback	1 (14)	1 (7)	0.59
Time to RTS, d, mean ± SD	39.0 ± 20.4	189.6 ± 123.2	0.005
Preinjury games played per season, mean ± SD	8.6 ± 1.7	13.5 ± 2.0	<0.001
Postinjury games played per season, mean ± SD	8.6 ± 4.6 [a]	11.2 ± 5.9	0.32

[a]Includes NFL games played post-NCAA participation

All athletes (100%) were able to return to sport (RTS) at their respective level of competition following surgery. The mean time from surgery to the first game played postoperatively was 139.4 ± 123.6 days, with a range of 12 to 427 days. The mean time to RTS for NCAA players was 39.0 ± 20.4 days, while NFL players took an average of 189.6 ± 123.2 days (P = 0.005; Figure 2).

Figure 2. Time to Return to sport (RTS) between National Collegiate Athletics Association (NCAA) Football and National Football League (NFL) Players Following Suture-Button Fixation.

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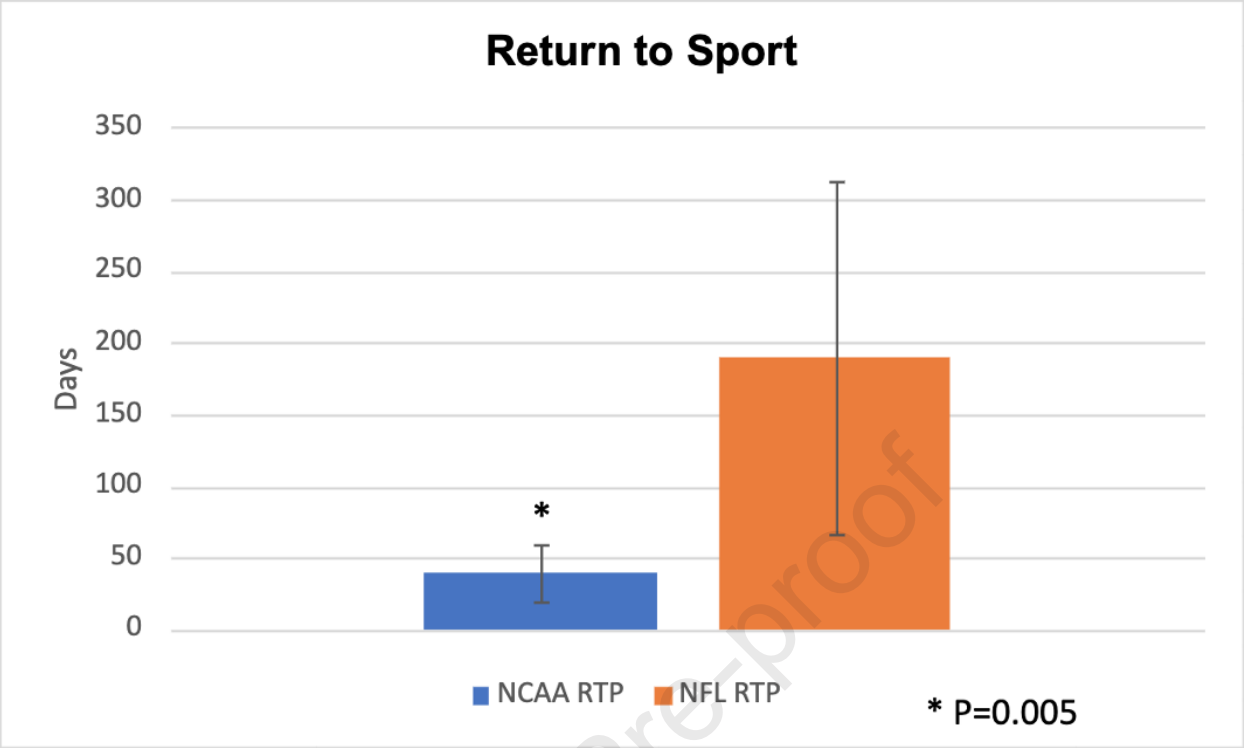

Following suture-button fixation, players participated in a mean of 23.48 ± 21.9 total games (range from 3-72). When compared to number of games played preoperativley, there was no statistically significant difference in the number of games played postoperatively between NCAA and NFL players (Figure 3).

Figure 3. Cumulative level of function trend in games played per season before and after suture-button fixation of syndesmotic ankle injuries between National Collegiate Athletics Association (NCAA) football and National Football League (NFL) players.

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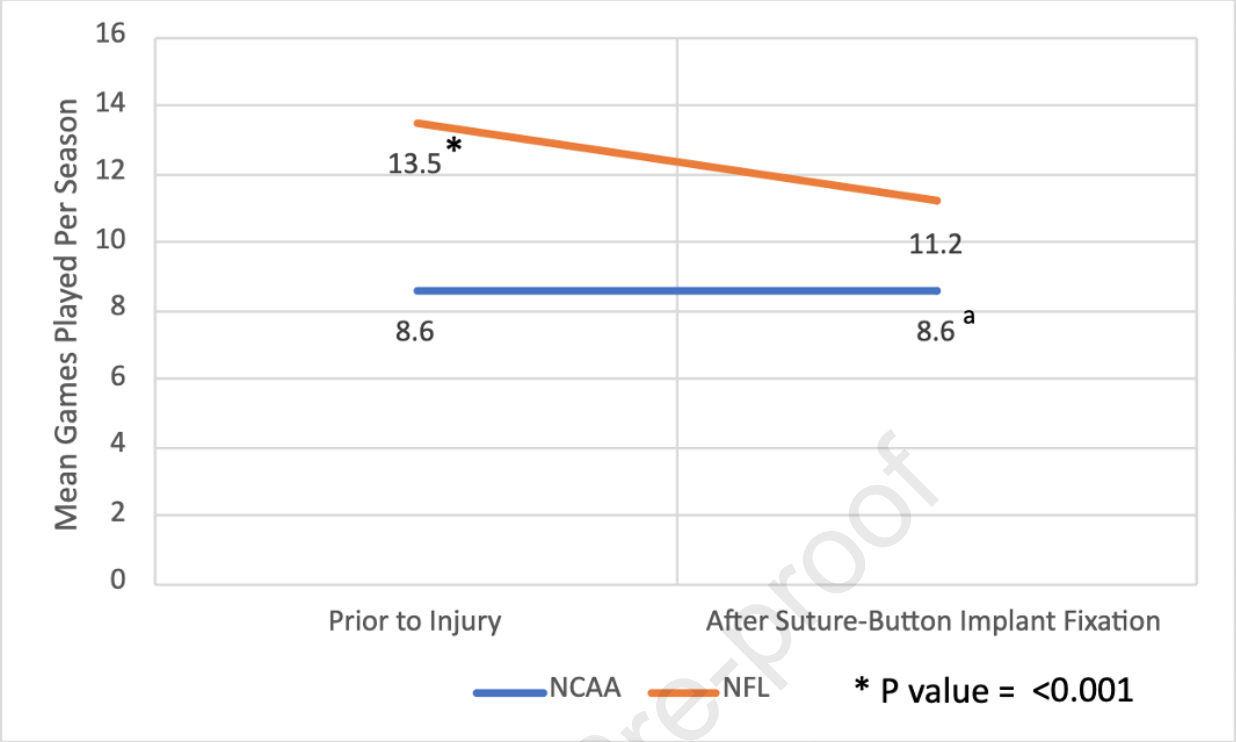

[a]Includes NFL games played post-NCAA participation
*A significant difference exists in number of preinjury games per season between NCAA and NFL players

Discussion

The results from this study underscore the effectiveness of suture-button fixation in managing unstable tibiofibular syndesmotic injuries among athletes, with a noteworthy achievement of 100% return-to-play (RTS) across both NCAA and NFL players. Previous studies have demonstrated similar results in elite level athletes who have undergone surgical intervention for syndesmotic injury with a RTS of 100%[19,20]. This outcome highlights the suture-button technique as a reliable surgical intervention for athletes requiring high levels of functional recovery as previously reported[12,13,17,18,21].

ACL reconstructions are one of the most studied orthopedic procedures in athletes, with well-established RTS timelines, failure rates, and performance metrics. Comparing anterior cruciate ligament reconstruction (ACLR) to suture-button fixation highlights that suture-button

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fixation shows higher RTS rates and quicker return times, positioning it as a promising intervention in elite athletes. A study of 215 NFL players conducted by Mody et al.[22] demonstrated a much lower RTS rate following ACLR of only 55.8%. Additionally, the authors found that within the first 3 years postinjury, players played in fewer games (8.7 vs 13.7; P < .0001) whereas our study found no significant difference in postinjury games played. A systematic review also performed on NFL players who underwent ACLR found a RTS rate of only 67.2%[23] Furthermore, the mean time to return was around 350 days, compared to a mean time to return of 189.6 days in our study cohort.

The decreased RTS rate and increased mean time to return to sport compared to suture-button fixation has not only been shown following ACLR but with other lower extremity and upper extremity surgeries as well[24–27]. Aune et al.[24] previously found a RTS of 61% and mean time to play of over 250 days in a study of 77 NFL players after arthroscopic lateral partial meniscectomies. Robins et al.[25] examined RTS in NCAA football athletes who had underwent arthroscopic surgery for recurrent shoulder instability. 85.4% of players were able to RTS with those who played in a higher percentage of games before the injury being more likely to return to sport.

Salameh et al.[28] previously performed a meta-analysis on return to sport after syndesmotic injuries in athletes and found a RTS of 99%, similar to our study. However, pooled mean time to RTS for surgically treated patients (suture-button and syndesmotic screws) was lower at 70.94 days (95% CI 47.04, 94.85) compared to 139.4 days in our analysis. The authors did find a shorter mean time to RTS of 39.33 days (95% CI 28.78, 49.88) for nonsurgically treated subjects. The literature specific to RTS and performance following surgical fixation of syndeosmotic injury remains sparse, especially when examining high level football players such

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as collegiate and NFL players. D'Hooghe et al.[29] did find a return to sport of 103±28 days for professional football players in the Qatar Football Association following suture-button fixation. 95% of players were able to return to competitive play within 6 months after surgery.

The observed disparity in RTS timelines between NCAA and NFL players, where NCAA athletes returned to play more quickly, can be attributed to several factors inherent to each level of competition. NCAA players, typically younger and less exposed to the prolonged physical demands and extensive recovery protocols seen in the NFL, may benefit from quicker rehabilitation and less stringent post-operative restrictions. In the cohort study performed by D'Hooghe et al.[29], increasing age was correlated with longer rehabilitation times and longer RTS, supporting this hypothesis. The older age of NFL players compared to NCAA players may also mean that they have more concomitant previous injuries and/or surgeries for other conditions. Additionally, the structure of NCAA athletic programs may afford players more intensive and individualized post-surgical care compared to their NFL counterparts, potentially facilitating a swifter recovery. Lastly, our results showed a significantly lower amount of pre-injury games played by NCAA athletes compared to NFL players. Especially with the more rigorous and physical season of the NFL, players may be held out from games as a precaution for longer periods of time compared to NCAA football players. There also may be a push for NCAA players to RTS sooner due to the fact that prolonged absence could affect their future career in the NFL, sponsorships, etc. Previous studies have found similar results following ACLR with a quicker mean time to return to sport for NCAA football players when compared to NFL players[23,30].

The study found no difference in level of function post-surgery between the two groups which suggests that while the timeline for RTS may vary, the suture-button fixation does not

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impact the capacity of athletes to handle their respective workloads upon return. Both NCAA and NFL players were able to resume their pre-injury activity levels, which indicates that the surgical technique does not compromise athletic performance or functional recovery. Injuries unrelated to the operative extremity did not interfere with the athletes' ability to return to sport. Thus, the primary concern for recovery was the ankle injury itself, and other potential injuries did not affect the athletes' capacity to resume competitive play at their pre-injury level. These findings have practical implications for clinical practice and rehabilitation strategies. For medical professionals treating tibiofibular syndesmotic injuries in athletes, the suture-button fixation method appears to offer a robust solution with favorable outcomes in terms of RTS such as high success rates, shorter recovery timelines (especially for NCAA players), and sustained performance capacity, all of which indicate the procedure was effective in enabling a full athletic comeback. However, the variation in RTS timelines between NCAA and NFL players suggests that individual factors such as age, level of competition, off-season program structure, and rehabilitation resources should be considered when planning post-operative care but this study was not able to draw a significant conclusion for these variables and warrants further investigation of these specific variables as they relate to RTS. There is no consensus regarding structured rehabilitation following surgical fixation of a syndesmotic ankle injury; however, the patient is typically kept non-weightbearing for a week followed by progressive weightbearing, increase in weight bearing exercises at 4 weeks, and more sport specific exercises/drills at 8+ weeks[31,32].Tailoring rehabilitation protocols to accommodate these differences could further optimize recovery and enhance the likelihood of a successful return to competitive play though due to the design of this study it was not possible to control or test for differences in rehabilitation.

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Limitations

This study is not without limitations. First, our sample size was small, which limits the power of our conclusions. It is possible that there are NFL or NCAA football players who were not included in the study due to inaccuracies in injury reports or difficulty in finding online sources documenting the injury. Additionally, although only reputable online sources were used to find information on players' injuries, time of surgery, and demographic at time of injury, it is possible that inaccuracies within the source were present. This method of finding player information also causes inherent selection bias as the authors are doing their own searching rather than using unavailable standardized NFL and NCAA medical records. Furthermore, the study did not compare individual performance statistics (e.g., yards gained, tackles made, or playing time) Patient-Reported Outcome Measures (PROMs), or Ankle activity score before and after injury, limiting the ability to evaluate whether athletes returned to their pre-injury level of performance/function. Overall, the results demonstrate that athletes, regardless of whether they were NCAA or NFL players, were able to successfully return to their respective sports after ankle surgery. While NCAA players returned to play significantly faster, both groups were able to achieve similar performance levels postoperatively in terms of the number of games played.

Conclusion

In this study, we found that athletes, regardless of whether they were NCAA or NFL players, were able to successfully return to their respective sports after ankle surgery. While NCAA players returned to play significantly faster, both groups were able to achieve similar performance levels postoperatively in terms of the number of games played.

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